

Final Exam

Macroeconomics II

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Problem 1 (Questions 1–20): Measurement, IS–LM, AD–AS

- ① Units. If CPI increases from 180 to 189 over a year, inflation is closest to B
- (A) 2%
 - (B) 5%
 - (C) 9%
 - (D) –5%

Explanations.

- (A) Wrong: too small.
- (B) Correct: $(189 - 180)/180 = 0.05$.
- (C) Wrong: 9% would be $180 \rightarrow 196.2$.
- (D) Wrong sign.

- ② Approximation. For small changes, $\log(P_t/P_{t-1})$ is approximately B
- (A) $P_t - P_{t-1}$
 - (B) $(P_t - P_{t-1})/P_{t-1}$
 - (C) $(P_t - P_{t-1})/P_t$
 - (D) P_{t-1}/P_t

Explanations.

- (A) Wrong: not scale-free.
- (B) Correct: first-order approximation.
- (C) Wrong denominator.

(D) Wrong.

③ Fisher (numbers). If $i = 1.5\%$ and $\mathbb{E}\pi = 2.0\%$, the ex-ante real rate is C

- (A) 3.5%
- (B) 0.5%
- (C) -0.5%
- (D) 2.0%

Explanations.

- (A) Wrong: adds.
- (B) Wrong sign.
- (C) Correct: $1.5 - 2.0 = -0.5$.
- (D) Wrong: equals expected inflation.

④ ZLB. If $i = 0$ and expected inflation falls from 1% to -1% , the real rate changes by C

- (A) -2 pp
- (B) 0 pp
- (C) $+2$ pp
- (D) $+1$ pp

Explanations.

- (A) Wrong sign.
- (B) Wrong: $r = -\mathbb{E}\pi$ at ZLB.
- (C) Correct: real rate rises by 2 percentage points.
- (D) Wrong.

⑤ Debt valuation view. The price level tends to rise when expected future primary surpluses B

- (A) increase
- (B) fall
- (C) stay constant
- (D) equal current taxes

Explanations.

- (A) Wrong: more backing raises real value.
- (B) Correct: weaker backing raises P to revalue nominal claims.
- (C) Not necessarily.
- (D) Not the key condition.

⑥ Okun. If $u - u^n = -0.4x$ and $x = -3\%$, then $u - u^n$ is B

- (A) -1.2%
- (B) $+1.2\%$
- (C) -0.75%
- (D) $+0.75\%$

Explanations.

- (A) Wrong sign.
- (B) Correct: $-0.4 \times (-3) = +1.2$.
- (C) Wrong.
- (D) Wrong.

⑦ Money market numerics. If $M/P = 12$, $k = 0.8$, $h = 4$, and $Y = 20$, then i equals A

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Explanations.

- (A) Correct: $i = (k/h)Y - (1/h)(M/P) = (0.2)20 - 0.25 \cdot 12 = 1$.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

⑧ IS numerics. If $Y = 200 - 50r$ and r falls by 0.01, then $\Delta Y =$ B

- (A) -0.5
- (B) $+0.5$
- (C) $+5$
- (D) -5

Explanations.

- (A) Wrong sign.
- (B) Correct: $-50 \times (-0.01) = +0.5$.
- (C) Wrong magnitude.
- (D) Wrong sign/magnitude.

⑨ AD slope (IS–LM). With M fixed, $P \uparrow$ leads to $Y \downarrow$ because B

- (A) real balances rise
- (B) real balances fall and i rises
- (C) IS shifts left mechanically
- (D) Y^{FE} rises

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑩ AD–AS numerics. AD: $y = 50 - 2p$, SRAS: $p = 1 + 0.5(y - 40)$. Equilibrium output y is C

- (A) 40
- (B) 42
- (C) 44
- (D) 46

Explanations.

- (A) Wrong.
- (B) Wrong.
- (C) Correct: solve to get $y = 44$.
- (D) Wrong.

⑪ Using the previous AD–AS system, the equilibrium price level p is C

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Explanations.

- (A) Wrong.
- (B) Wrong.
- (C) Correct: $p = 0.5 \cdot 44 - 19 = 3$.
- (D) Wrong.

⑫ Diagnosis. A positive demand shock in AD–AS typically causes A

- (A) $y \uparrow, p \uparrow$
- (B) $y \downarrow, p \uparrow$
- (C) $y \uparrow, p \downarrow$
- (D) $y \downarrow, p \downarrow$

Explanations.

- (A) Correct.
- (B) Supply shock pattern.
- (C) Wrong.
- (D) Wrong.

⑬ **Diagnosis.** A negative supply shock in AD–AS typically causes B

- (A) $y \uparrow, p \uparrow$
- (B) $y \downarrow, p \uparrow$
- (C) $y \uparrow, p \downarrow$
- (D) $y \downarrow, p \downarrow$

Explanations.

- (A) Wrong.
- (B) Correct (stagflation).
- (C) Wrong.
- (D) Wrong.

⑭ **Medium run.** Output returns toward y^{FE} mainly because B

- (A) AD shifts back automatically
- (B) expected prices/wages adjust (SRAS shifts)
- (C) money supply is constant
- (D) productivity always rises

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

①⑤ IS–LM: a cut in lump-sum taxes (holding G fixed) is typically modeled as A

- (A) IS shifts right
- (B) LM shifts right
- (C) IS shifts left
- (D) LM shifts left

Explanations.

- (A) Correct: autonomous demand rises.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

①⑥ IS–LM: a decrease in nominal money M (holding P fixed) shifts B

- (A) LM right/down
- (B) LM left/up
- (C) IS right
- (D) IS left

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

①⑦ Liquidity trap logic. If the money-market-clearing formula implies $i^* < 0$ and there is a ZLB, then the equilibrium i is B

- (A) i^*
- (B) 0
- (C) $|i^*|$
- (D) undetermined always

Explanations.

- (A) Wrong: violates ZLB.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

⑱ One-time jump. If P rises by 6% once and then stays constant, inflation is B

- (A) 6% forever
- (B) 6% once then 0%
- (C) 0% forever
- (D) -6% once then 0%

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong sign.

⑲ Anchored expectations means A

- (A) $\mathbb{E}\pi$ stays near target after temporary shocks
- (B) $\pi = 0$ always
- (C) $x = 0$ always
- (D) i never changes

Explanations.

- (A) Correct.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

②0 IS–MP–PC uses which three blocks? B

- (A) IS, LM, Phillips
- (B) IS, policy rule, Phillips
- (C) AD, money demand, LRAS
- (D) budget constraint, money demand, Phillips

Explanations.

- (A) Wrong: LM replaced by a rule.
 - (B) Correct.
 - (C) Wrong.
 - (D) Wrong.
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Problem 2 (Questions 21–40): IS–MP–PC, Taylor principle, NK

②1 In Whelan’ s IS–MP derivation, the key term is $(\beta_{\pi} - 1)$ because the real-rate gap contains A

- (A) $i - \pi$
- (B) $i + \pi$
- (C) $\pi - i$
- (D) i/π

Explanations.

- (A) Correct: $r = i - \pi$ (or $i - \pi^e$).
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

(22) Taylor principle. $\beta_\pi > 1$ ensures that when inflation rises, the real rate B

- (A) falls
- (B) rises
- (C) is unchanged
- (D) must be zero

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(23) If $\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$ and $\beta_\pi > 1$, then C

- (A) $\theta > 1$
- (B) $\theta = 1$
- (C) $0 < \theta < 1$
- (D) $\theta < 0$

Explanations.

- (A) Wrong.
- (B) Wrong.
- (C) Correct.
- (D) Wrong.

(24) Numerics. If $\alpha = 1.2$, $\gamma = 0.4$, $\beta_\pi = 2$, then θ is closest to A

- (A) 0.68
- (B) 0.32
- (C) 1.68
- (D) 0

Explanations.

- (A) Correct: denom $1 + 1.2 \cdot 0.4 \cdot 1 = 1.48$.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

②5 A tougher central bank corresponds to B

- (A) lower β_π
- (B) higher β_π
- (C) higher γ
- (D) lower α

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong: γ is Phillips slope.
- (D) Not the toughness definition.

②6 With a tougher CB, a cost-push shock tends to produce B

- (A) larger inflation rise, smaller output fall
- (B) smaller inflation rise, larger output fall
- (C) smaller inflation rise, smaller output fall
- (D) no changes

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(27) Explosive region in the TaylorRule discussion: if $1 - \frac{1}{\alpha\gamma} < \beta_\pi < 1$, then B

- (A) IS–MP slopes down and is stable
- (B) IS–MP slopes up and can be unstable under adaptive expectations
- (C) inflation is pinned at target
- (D) Phillips curve becomes vertical

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

(28) Boundary. If $\alpha\gamma = 0.5$, then $1 - \frac{1}{\alpha\gamma} =$ B

- (A) 0.5
- (B) -1
- (C) 1
- (D) 2

Explanations.

- (A) Wrong.
- (B) Correct: $1 - 2 = -1$.
- (C) Wrong.
- (D) Wrong.

(29) Two-period NK IS (calculation). With $x_1 = 0$, $\sigma = 1$, $r_0^n = 1\%$, $i_0 = 4\%$, and $\mathbb{E}_0\pi_1 = 2\%$, x_0 is C

- (A) $+1\%$
- (B) 0%
- (C) -1%
- (D) -3%

Explanations.

- (A) Wrong sign.
- (B) Wrong.
- (C) Correct: $4 - 2 - 1 = 1$ so $x_0 = -1\%$.
- (D) Wrong.

30 Forward-looking NKPC is B

- (A) $\pi_0 = \pi_{-1} + \kappa x_0 + u_0$
- (B) $\pi_0 = \beta \mathbb{E}_0 \pi_1 + \kappa x_0 + u_0$
- (C) $\pi_0 = \kappa x_0$ always
- (D) $\pi_0 = \beta \pi_0 + \kappa x_0$

Explanations.

- (A) Backward-looking.
- (B) Correct.
- (C) Only if expectations and shocks are zero.
- (D) Wrong.

31 NKPC numerics. If $\beta = 0.99$, $\mathbb{E}_0 \pi_1 = 1\%$, $\kappa = 0.3$, $x_0 = 2\%$, $u_0 = 0$, then π_0 is A

- (A) 1.59%
- (B) 1.29%
- (C) 0.99%
- (D) 0.60%

Explanations.

- (A) Correct: $0.99 + 0.6 = 1.59$.
- (B) Wrong.
- (C) Wrong: ignores gap term.
- (D) Wrong: gap term only.

32) Taylor rule numerics. If $i_0 = \bar{i} + \phi_\pi \pi_0$ with $\bar{i} = 1\%$, $\phi_\pi = 1.5$, and $\pi_0 = 3\%$, then i_0 is C

- (A) 3%
- (B) 4.5%
- (C) 5.5%
- (D) 6%

Explanations.

- (A) Wrong.
- (B) Wrong (missing intercept).
- (C) Correct: $1 + 4.5 = 5.5$.
- (D) Wrong.

33) Two-period Fisher numerics. If $i_0 = 0\%$ and $\mathbb{E}_0 \pi_1 = 2\%$, then r_0 is C

- (A) 2%
- (B) 0%
- (C) -2%
- (D) -4%

Explanations.

- (A) Wrong sign.
- (B) Wrong.
- (C) Correct.
- (D) Wrong.

34) Two-period IS sign. Holding $\mathbb{E}_0 \pi_1$ fixed, a decrease in i_0 implies B

- (A) x_0 decreases
- (B) x_0 increases
- (C) x_0 unchanged
- (D) π_0 falls one-for-one

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

35 In NKPC, increasing u_0 is best interpreted as B

- (A) a right shift in IS
- (B) an upward shift in inflation for given x_0 and expectations
- (C) a fall in r_0^n
- (D) a decrease in κ

Explanations.

- (A) Wrong.
- (B) Correct.
- (C) Wrong.
- (D) Wrong.

36 Two-period closure. If $x_1 = 0$ and $\pi_1 = 0$ are imposed, then at $t = 0$ A

- (A) there are no expectations terms
- (B) the system is purely backward-looking
- (C) policy becomes irrelevant
- (D) the LM curve returns

Explanations.

- (A) Correct: $\mathbb{E}_0 x_1 = \mathbb{E}_0 \pi_1 = 0$.
- (B) Wrong.
- (C) Wrong.
- (D) Wrong.

37) Determinacy in basic NK. A standard sufficient condition is B

- (A) $\phi_\pi < 1$
- (B) $\phi_\pi > 1$
- (C) $\kappa = 0$
- (D) $\sigma = 0$

Explanations.

- (A) Wrong.
- (B) Correct (Taylor principle).
- (C) Wrong.
- (D) Wrong.

38) Two-period NK IS (calculation). Suppose $x_1 = 0$, $\sigma = 1$, $r_0^n = 1\%$, $i_0 = 3\%$, and $\mathbb{E}_0\pi_1 = 0\%$. Then x_0 is B

- (A) $+2\%$
- (B) -2%
- (C) -1%
- (D) 0%

Explanations.

- (A) Wrong sign.
- (B) Correct: $3 - 0 - 1 = 2$ so $x_0 = -2\%$.
- (C) Wrong.
- (D) Wrong.

39) IS-MP (calculation). If $\alpha = 1.5$ and $\beta_\pi = 0.8$, then $-\alpha(\beta_\pi - 1)$ equals B

- (A) -0.3
- (B) $+0.3$
- (C) -1.2
- (D) $+1.2$

Explanations.

- (A) Wrong sign.
- (B) Correct: $-1.5(-0.2) = +0.3$.
- (C) Wrong magnitude.
- (D) Wrong magnitude.

④0 IS–MP slope (sign). If $\beta_\pi = 0.6$ and $\alpha > 0$, then IS–MP in (y, π) space is B

- (A) downward sloping
- (B) upward sloping
- (C) vertical
- (D) flat

Explanations.

- (A) Wrong: downward requires $\beta_\pi > 1$.
- (B) Correct: $\beta_\pi < 1$ implies positive slope.
- (C) Wrong.
- (D) Wrong.